trignometric integrals

- these are integrals that have trignometric Function in them and you can evaluate them by u-substitution or using trig-identites to Solve

Quick review

$\tan\theta = \frac{\sin\theta}{\cos\theta}$ or $\frac{1}{\cot\theta}$

$\sec\theta = \frac{1}{\cos\theta}$

$\csc\theta = \frac{1}{\sin\theta}$

$\cos\theta = \frac{1}{\sec\theta}$

trig-identites

- $\sin^2\theta + \cos^2\theta = 1$
- $\tan^2\theta + 1 = \sec^2\theta$
- $\cot^2\theta + 1 = \csc^2\theta$
- $\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos(2\theta)$
- $\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos(2\theta)$

- You can use all of these to replace terms in the integral to make it easier to to Solve

☆ because these are indefinte integrals you need to add a plus "C" to the end

Practice Problems

1) $\int \sin^2(x)\,dx$

$\sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos(2\theta)$

$\int \frac{1}{2} - \frac{1}{2}\cos(2x) \Rightarrow \int \frac{1}{2} - \frac{1}{2}\int \cos(2\theta)$

$u = 2\theta$

$du = 2\,d\theta$

$\frac{du}{2} = d\theta$

$\int \frac{1}{2} - \frac{1}{2}\int \cos(2\theta)\,d\theta$

$\frac{1}{2}\theta - \frac{1}{2}\int \cos(u)\cdot\frac{du}{2}$

$\frac{1}{2}\theta - \frac{1}{4}\left[\sin(2\theta)\right]$

$\Rightarrow \boxed{\frac{1}{2}\theta - \frac{1}{4}\sin(2\theta)}$

- this is not in terms of "x" its in terms of "θ"

2) $\int \sin(x)\cos(x)\cdot dx$

$u = \sin(x)$

$du = \cos(x)\,dx$

$\frac{du}{\cos(x)} = dx$

$\int u \cdot \cancel{\cos(x)} \cdot \frac{du}{\cancel{\cos(x)}}$

$\int u \cdot du \approx> \left[\frac{(\sin(x))^2}{2}\right]$

$\boxed{\frac{(\sin(x))^2}{2}}$

3) $\int \sec^2(\theta)\,d\theta$

- this one should be memorized this was just an example of how you can use trig-identites

$\tan^2\theta + 1 = \sec^2\theta$

$\int \tan^2\theta + 1 \; d\theta$

$\Rightarrow \left[\tan(\theta) - \cancel{\theta} + \cancel{\theta}\right] \approx> \boxed{\tan(\theta)}$

4) $\int \sin^3(x) \Rightarrow \int \sin^2(x)\cdot \sin(x)$

$\sin^2(x)+\cos^2(x)=1$

$\sin^2(x)=1-\cos^2(x)$

$\int (1-\cos^2(x))\cdot \sin(x)$

$\int \sin(x)-\sin(x)\cdot \cos^2(x)\,dx$

$\int \sin(x)\,dx-\int \sin(x)\cos^2(x)\,dx$

$\cos(x)-\int \sin(x)\cos^2(x)\,dx$

$u=\cos(x)$

$du=\sin(x)\,dx$

$\frac{du}{\sin(x)}=dx$

$\cos(x)-\int \cancel{\sin(x)}\cdot (u)^2\cdot \frac{du}{\cancel{\sin(x)}}$

$\cos(x)-\int (u)^2\cdot du$

$$\boxed{\cos(x)-\frac{(\cos(x))^3}{3}+c}$$

5) $\int \sin^2(x)\cos(x)\,dx$

$u=\sin(x)$

$du=\cos(x)\,dx$

$\frac{du}{\cos(x)}=dx$

$\int (u)^2\cdot \cancel{\cos(x)}\cdot \frac{du}{\cancel{\cos(x)}}$

$\int (u)^2\cdot du$

$$\boxed{\frac{(\sin(x))^3}{3}+c}$$